7. Project Demonstration Phase

7.1 Introduction

The demonstration phase presents the completed LegalEase application and shows its major functionalities in a real-world usage scenario. It begins with registration and login and continues through document selection, data entry, draft generation, clause explanation, review, editing, and export.

7.2 Objectives of Demonstration

- To demonstrate the overall functionality of LegalEase.
- To show the user authentication system.
- To demonstrate AI draft generation from guided inputs.
- To demonstrate plain-language clause explanation and document review.
- To demonstrate editing, clause regeneration, and export.
- To display the history of saved documents.
- To show the interaction between frontend, backend, database, and Gemini models.
- To present the final working application to the evaluator.

7.3 Demonstration Environment

The demonstration is performed in the configured development environment on a laptop with an internet connection, using VS Code to run the FastAPI server, a web browser for the frontend, the configured database, and a valid Google Gemini API key.

7.4 Demonstration Workflow

Step	Feature	Demonstration
1	Registration & Login	Create an account and log in securely.
2	Document Selection	Choose a document type and jurisdiction.
3	Questionnaire	Enter party details, dates, amounts, and key terms.
4	Draft Generation	Generate the draft and show the numbered clauses.
5	Clause Explanation	Select a clause and view the plain-language explanation.
6	Review	Run the review and show missing clauses and risk suggestions.
7	Editing	Edit a clause and regenerate another with extra instructions.
8	Export	Preview the document and export it as PDF and Word.
9	History	Open previously saved documents.
10	Logout	End the session.

7.5 Sample Scenario

A freelance designer needs a service agreement for a new client. The designer selects the freelance contract type, enters both parties' names, the project scope, the payment amount and schedule, the delivery date, and the jurisdiction. LegalEase generates a draft with scope, payment, intellectual property, confidentiality, termination, and dispute resolution clauses. The designer asks for the termination clause to be explained in simple words, runs the review, and adds a late-payment clause

suggested by it. The designer then exports the agreement as a PDF to share with the client, with a note that it should be checked by a lawyer before signing.

7.6 Results and Observations

- The application meets the defined functional requirements.
- Gemini produces well-structured drafts that follow the entered details.
- Plain-language explanations make the clauses easier to understand.
- The review feature helps identify missing clauses and unclear wording.
- Exported documents are neatly formatted and readable.
- The interface is simple and easy to navigate.
- User data is protected through authentication.

7.7 Limitations

- Drafts are for information only and are not legal advice; they should be reviewed by a qualified lawyer.
- AI output may contain mistakes or may not reflect the latest law of a particular jurisdiction.
- An internet connection and Gemini API access are required.
- Response time depends on API availability and network speed.
- Only a limited set of document types is supported in the current version.
- Generated content may be subject to safety filters.

7.8 Future Enhancements

- More document types and jurisdiction-specific templates.
- Upload and analysis of existing contracts.
- Support for multiple languages.
- Electronic signature and sharing with the other party.
- Lawyer review and consultation option.
- Mobile application.

7.9 Conclusion

LegalEase: AI-Powered Legal Document Generator combines guided data collection, AI drafting, plain-language clause explanation, document review, editing, and export in a single application. The demonstration confirms that the system works as intended and gives users a convenient way to prepare clear, well-structured legal document drafts, while reminding them to seek professional review before relying on them.